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UNIVERSITÄT
HEIDELBERG
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SEIT 1386

AI in research software: Best practices

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February 2026



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1. Requirements of “ML-based science”

What this course is not

- An introduction to data science
- An introduction to machine learning
- A course about different ML algorithms
- A course about different ML training approaches and libraries
- ...

What this course is

- A best practices guide to creating machine learning based research software (MLBRS)
- A recommendation on how to manage and prepare your data
- A recommendation on how to train your models
- An introduction to software engineering best practices for MLBRS
- A guideline on how to generate independently reproducible scientific results using data-based approaches
- A guideline on how to publish your data and your models in an interoperable and responsible way

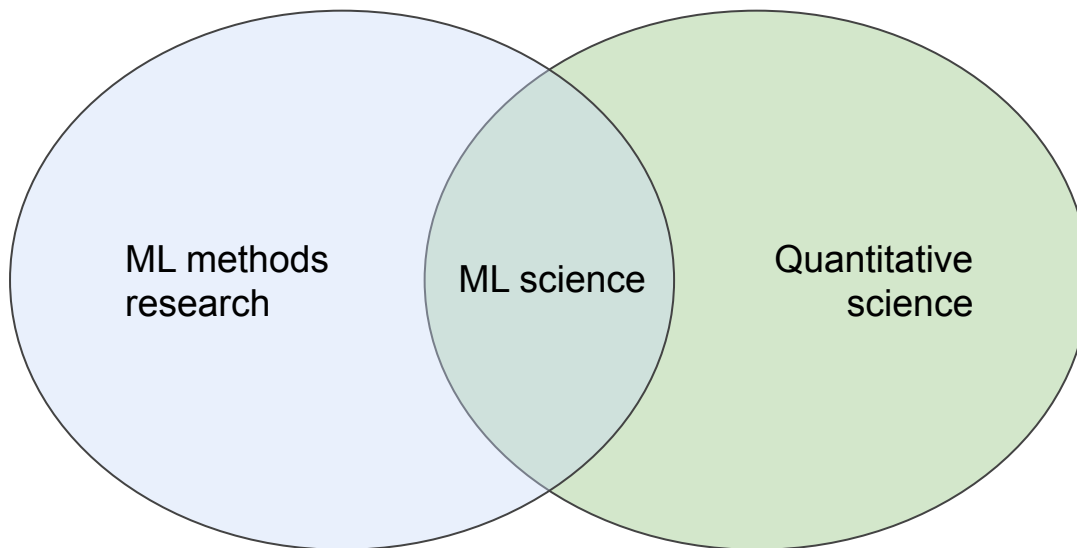
What is special about research software
based on data? (“ML-based science”)

ML science

- Scientific research that uses machine learning models to extend scientific knowledge
- Answers a scientific question by using ML
- No restriction on algorithm, method, library, domain

Contrary to:

- ML methods research:
Research on ML methods and algorithms with the goal to improve the field of ML



Research software

“... software that is developed and used in the context of research...”

Shifting requirements

A scientific question is answered using computation/simulation, but the way the problem is solved changes as part of the research process.

Passed along researchers

Initially developed for one purpose but then often organically extended depending on the researcher's needs.

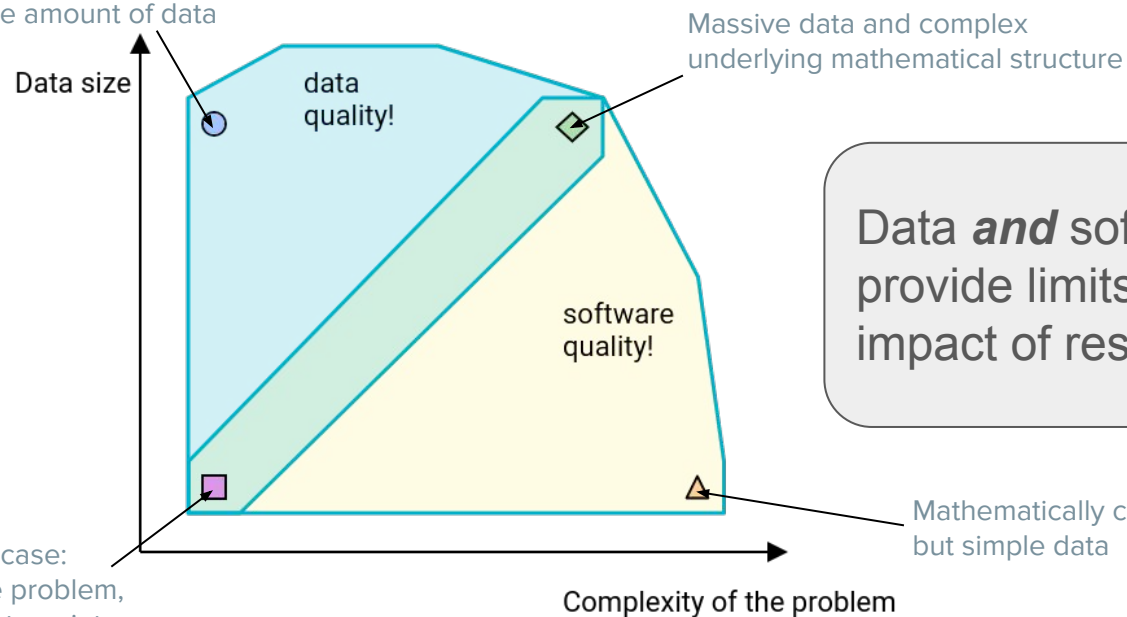
Development Practices

Often created by researchers that have no fundamental training in software engineering and inherit practices from those around them.

ML-based research software

“... software that is developed and used in the context of research and predicts outcomes based on initial data...”

Mathematically simpler problem,
but large amount of data



Trivial case:
simple problem,
few data points

Data **and** software quality
provide limits to quality and
impact of research!

Mathematically complex problem,
but simple data

MLBRS: Data

Data is foundation for..

...model training, decision making and/or predictions.

Different kinds of data

For example, numerical data, textual data, images, audio, video.

Metadata

What is relevant metadata and should be included on the data card?

Availability and licensing

Will the data be publicly available to the community? What license does/will the dataset have?

Legal considerations

Where does the data come from? Is it licensed? Is it public or private data? In what form is the data stored and processed?

Ethical considerations

Does the data exploit work of others? Does it break some sort of confidentiality? Will it impact in a possible harmful way or can it be misconstrued to do harm?

Bias

Is there an inherent bias in the data itself, due to the data collection approach, or other reasons?

MLBRS: Software

Purpose

Will the software be more widely used, be an in-house code, or one-person software?

Software engineering best practices

Does the software follow software engineering best practices (version control, testing, documentation, ...)?

Usability and reproducibility

Does the software include documentation on how models can be trained, and keeps track of training parameters? Does the software help to generate model cards and provide models in transferable format?

Accuracy and reliability

Does the software create robust and consistent results, even though it is based on a non-deterministic process?

Legal considerations

Does the software incorporate third-party models and/or code?

Legal considerations

*What license is the software published under?
What license are models published under?*

Security

Is the software secure against data injection?

Reproducibility

- Provide data to enable others to reproduce findings
- Provide code to enable others to reproduce findings

→ ***Computational reproducibility (i)***

- Make sure your findings are true findings, and do not arise from problems with your data/code

→ ***Independent reproducibility (ii) // “Responsible AI”***

Research software engineering generally targets (i), but with MLBRS we target (ii)

Why should you care?

Your research integrity, scientific best conduct (malpractice), can have long-lasting detrimental effect on science (impact on others and the field), affects society!

Key aspects

Legal aspects

Documentation on
model training,
hyperparameter
tuning, model
testing

Model bias

Software
quality

Reproducibility
of the model's
predictions

Robustness of
the model(s)

Software
security

Interoperability

Reproducibility
of the model
training

Ethical
aspects

Data bias

Responsible
AI

Documentation on data
collection,
data
cleaning,
feature
selection

Data leakage

